Yunhan Qiao

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Research Interests

My research sits at the intersection of empirical software engineering, code comprehension, and human–AI collaboration. I study how students and experienced developers work with GenAI coding assistants and agents (e.g., GitHub Copilot, Cursor) in realistic *brownfield* codebases, and whether faster, more correct work also means deeper understanding. I combine controlled experiments, screen-recorded behavioral analysis, prompt coding, EEG, and interviews to inform how computing education teaches and assesses GenAI-assisted development and how future AI coding agents and IDEs should be designed.

Education

- **Oregon State University** — Corvallis, OR, United States
  Ph.D. in Computer Science, School of EECS — *Sep 2023 – Jun 2027 (expected)*
  - **Advisor**: Dr. Christopher Hundhausen at the SEAL Lab
  - **Research Areas**: Empirical Software Engineering, Code Comprehension, Computing Education, Human–AI Interaction
- **Arizona State University** — Tempe, AZ, United States
  M.S. in Software Engineering — *Aug 2021 – May 2023*
  - **Advisor**: Dr. Robert LiKamWa
  - **Thesis**: Networked System for Volumetric Athletic Coaching in Augmented Reality
- **Guangxi University of Science and Technology** — Liuzhou, China
  B.E. in Software Engineering — *Sep 2016 – Jun 2020*
- **Southern Cross University** — Gold Coast, Australia
  B.E. in Information Technology — *Sep 2017 – Jun 2020*

Publications

Journal Articles & Conference Papers

- **Yunhan Qiao**, Md Istiak Hossain Shihab, and Christopher Hundhausen. **A Systematic Literature Review of the Use of GenAI Assistants for Code Comprehension: Implications for Computing Education Research and Practice.** ***TOCE****: ACM Transactions on Computing Education*, 26(2), Article 24, 1–33, 2026. [PDF] [DOI]
  Selected by the TOCE editorial board for presentation at SIGCSE TS 2027.
- Ahsun Tariq, Phillip Conrad, Christopher Hundhausen, Sanjay Chandrasekaran, **Yunhan Qiao**, and Summit Haque. **Using SlackBot-Prompted Reflections to Promote Contribution Equality in Team Software Development Projects.** ***TOCE****: ACM Transactions on Computing Education*, 2026. [PDF] [DOI]
- Md Istiak Hossain Shihab, Christopher Hundhausen, Ahsun Tariq, Summit Haque, **Yunhan Qiao**, and Brian Wise Mulanda. **The Effects of GitHub Copilot on Computing Students' Programming Effectiveness, Efficiency, and Processes in Brownfield Coding Tasks.** ***ICER'25****: Proceedings of the 2025 ACM Conference on International Computing Education Research*, Vol. 1, 407–420, Aug. 2025. [PDF] [DOI]
- Manish Motwani, Aakash Kulkarni, **Yunhan Qiao**, Matthew Davis, and Ziyan Chen. **LLM-Guided Differential Fuzzing for Detecting Platform-Specific Bugs in Scientific Applications.** ***AIxSE'25****: IEEE International Conference on AI x Software Engineering*, 59–66, 2025. [PDF] [DOI]

Under Review

- **Yunhan Qiao**, Md Istiak Hossain Shihab, Summit Haque, and Christopher Hundhausen. **Faster but Not Wiser: GitHub Copilot Decouples Programming Performance from Code Comprehension in Brownfield Tasks.** ***TOCE****: ACM Transactions on Computing Education*, under review. [arXiv]
- **Yunhan Qiao**, Summit Haque, and Christopher Hundhausen. **Beyond the Prompt: Linking What Developers Ask, Do, and Understand with Coding Agents.** ***SANER'27****: IEEE International Conference on Software Analysis, Evolution and Reengineering*, under review. [arXiv]

Under Submission

- O. Schutz, **Yunhan Qiao**, Fatima A. Moussaoui, Rudrajit Choudhuri, and Anita Sarma. **Heart and Mind in the Loop: Why We Vibe When We Work with AI.** ***CHASE'27****: International Conference on Cooperative and Human Aspects of Software Engineering*, under submission.

Teaching Experience

- **Graduate Teaching Assistant, School of EECS** — Oregon State University
  Undergraduate software engineering courses — *2024 – current*
  - **CS 362**: Software Engineering II
  - **CS 461**: Senior Software Engineering Project
  - **CS 467**: Online Capstone Project

- **CS 492**: Mobile Software Development

Professional Experience

- **Software Engineering Intern, Mocha Software** Nanjing, China
  Software development *Aug 2020 – Nov 2020*
  - Developed a conference system for China Mobile Migu Culture & Technology Group with Spring, Spring MVC, and MyBatis, using Ajax to cut page rendering time from 1 s to 0.5 s and dynamic SQL with Redis-backed SSO sessions to reduce search load.

Awards & Scholarships

- **EECS Graduate Scholarship**: Oregon State University *2023 – current*
- **Merit Award**: Guangxi University of Science and Technology *2018*

Academic Service

- **Reviewer**: SIGCSE TS 2027, the 58th ACM Technical Symposium on Computer Science Education
- **Presenter**: SIGCSE TS 2027, journal-first presentation

Skills & Coursework

- **Research Methods**: Qualitative Analysis, Quantitative Analysis, Interviews, Survey Design
- **Programming**: Python, Java, C, JavaScript, HTML, CSS, SQL, Bash
- **Frameworks & Tools**: React, Express, Flask, MongoDB, PostgreSQL, Git, AWS, Docker, LaTeX, Figma; Claude Code, GitHub Copilot, Cursor, OpenAI Codex
- **Coursework**: Machine Learning; NLP with Deep Learning; Software Engineering for ML; Program Analysis; Algorithms & Data Structures; Human–Computer Interaction; Field Studies in SE & HCI; Methods of Data Analysis

Languages

- **English**: Fluent
- **Mandarin Chinese**: Native